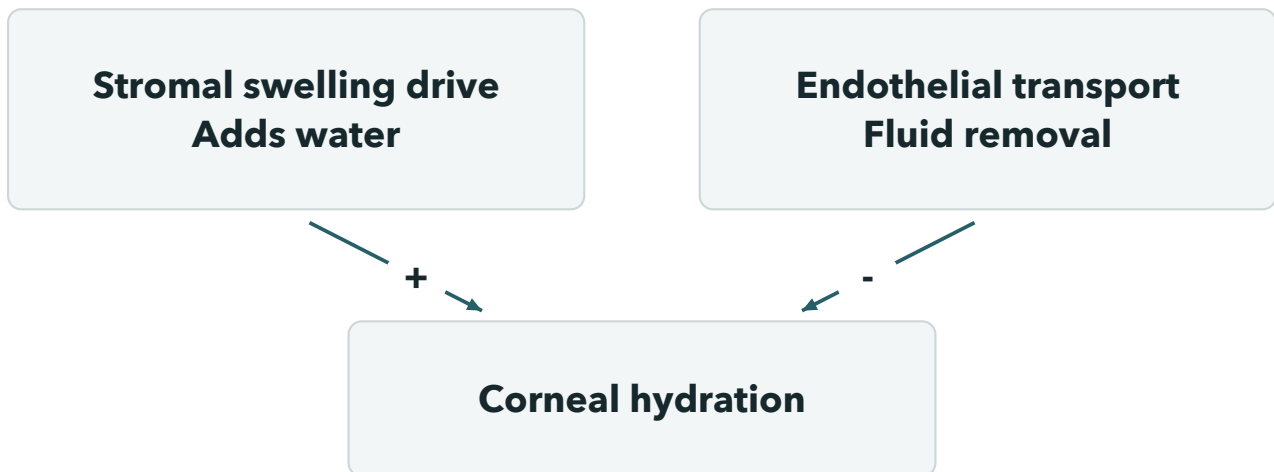


Corneal hydration and oxygen economy

Part 1 | Hydration is a dynamic balance

Similar corneal swelling can follow increased leak or reduced active removal.



Thickness reflects their balance, not one isolated process

Opposed stromal hydration and endothelial fluid-removal influences; arrows indicate functional balance.

Two experiments

Endothelial Na⁺/K⁺-ATPase inhibition reduces transport-dependent fluid removal. Increased stromal fixed-charge effects strengthen water uptake. Both thicken the cornea but act on different sides of the hydration balance; thickness alone does not locate the mechanism.

Do not reduce the pump to one ion

Bicarbonate supports transport and buffering; lactate handling also matters.

Metabolic and transport coupling

- Corneal metabolism produces lactate
↓
- Endothelial transport facilitates lactate efflux
↓
- Buffering supports this flux
↓
- Coupled osmotic processes support fluid removal

Interpret the experiment

Bicarbonate removal can impair pump function without destroying cells.

Na⁺/K⁺-ATPase supports gradients used by other transport pathways.

Rabbit, mouse and cultured-cell findings must retain their model context.

High-yield qualification

The established clinical concept is transport-dependent deturgescence. A diagram showing a single bicarbonate channel pumping all water would overstate a complex, evolving molecular model.

Permeability and transmissibility

Material oxygen permeability and whole-lens transport are different quantities.

Quantity

Feature	Expression	What changes it?
Permeability	Dk	Material diffusion and solubility
Transmissibility	Dk/t	Material and local lens thickness
Clinical oxygen exposure	Not Dk/t alone	Fit, tears and wearing conditions

Controlled comparison

- Hold polymer and conditions fixed
↓
- Halve lens thickness
↓
- Double Dk/t
↓
- Do not assume doubled clinical oxygen delivery

Calculation habit

Use matched thickness units. A thinner design raises Dk/t while material Dk stays constant.

Hypoxia changes metabolism

Reduced oxygen supply can add a metabolic swelling drive.

Physiological chain

- Oxygen supply falls
- ↓
- Anaerobic metabolism becomes more important
- ↓
- Lactate burden and local chemistry change
- ↓
- Corneal hydration and function may deteriorate

Clinical links

Closed-eye conditions differ from open-eye exposure.

A thick region of a lens may transport less oxygen locally.

Limbal tissue can be affected by chronic contact-lens stress.

Keep inference proportional

Oxygen transmissibility is informative, but one material number does not describe every location or wearing condition. Mechanical fit, surface disease and inflammation can coexist with hypoxic effects.

Reserve and a stress test

Resting clarity does not measure every part of endothelial functional reserve.

Observation

Feature	What it suggests	What it cannot prove
Clear cornea at baseline	Current balance adequate	Unlimited transport reserve
Swelling after a challenge	Balance was exceeded	A unique molecular defect
Recovery after removal	Some functional recovery	No structural injury

Use a controlled comparison

Keep bath pressure and osmolality constant.

Measure thickness over time rather than one endpoint.

Separate barrier integrity, viability and transport activity.

Reason from dynamics

A reversible pharmacological inhibition is useful precisely because it can alter function before changing cell number. Interpretation should connect the intervention to the measured response.